CampusX

CLASSES SPECIFICATIONS DOCUMENT

Peer-to-Peer College Student Marketplace

Item	Description
Purpose	Defines properties, behaviour, and relationships for principal classes and components.
Architecture	Clean Architecture: Domain, Application, Infrastructure, and WebUI.
Specification basis	CampusX SRS and the existing Classes Specifications baseline. Confirm unresolved assumptions with the team.
Table format	Every class/component uses a Relations column alongside Details / responsibility; multiplicity is stated where supported.

Design note: multiplicities shown here reflect the current specification baseline, not a confirmed database schema. Optional user reporting, soft deletion, and final status values must be confirmed.

Domain Layer — Entities

ApplicationUser

Responsibility: Represents a student or administrator profile.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	Id	Guid	Stable identifier mapped to the Identity record.	Referenced by Product.SellerId, Report.ReporterId/TargetUserId, and ContactInquiry.BuyerId/SellerId.
Properties	FullName	string	Required display name.	Used by profile and admin DTOs; no direct entity relationship.
Properties	Email	string	Required email; validated by identity workflow.	Mapped to the authentication identity record.
Properties	PhoneNumber	string?	Optional phone number.	No direct entity relationship.
Properties	CollegeName	string	Required college name.	Used for profile display/filtering if supported.
Properties	ProfileImageUrl	string?	Optional profile image URL.	No direct entity relationship.
Properties	CreatedAt	DateTime	UTC creation time; private set.	No direct entity relationship.
Properties	IsBlocked	bool	Account block state; false initially.	Checked by authorization/account workflows.
Properties	Role	UserRole	Student or Admin.	Used by authorization policies; enum relationship.
Behaviour	UpdateProfile(...)	Method	Updates editable profile fields only.	Does not change Id, CreatedAt, or Role.
Behaviour	BlockAccount() / UnblockAccount()	Methods	Change account block state.	Used by admin workflows through IAdminService.

Product

Responsibility: A product listing offered by a student.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	Id	Guid	Unique product identifier.	Referenced by Report.TargetProductId and ContactInquiry.ProductId.
Properties	Title	string	Required product title.	Displayed in product summary/details DTOs.
Properties	Description	string	Required listing description.	No direct entity relationship.
Properties	Price	decimal	Must be greater than zero.	No direct entity relationship.
Properties	ImageUrl	string	Product image URL; requirement to confirm.	No direct entity relationship.
Properties	Status	ProductStatus	Available, Sold, Removed, or Archived (confirm final values).	Uses ProductStatus enum.
Properties	SellerId	Guid	Owner's user identifier.	Many Products belong to one ApplicationUser: ApplicationUser 1 : 0..* Product.
Properties	CategoryId	Guid	Product category identifier.	Many Products belong to one Category: Category 1 : 0..* Product.
Properties	CreatedAt / UpdatedAt	DateTime	Creation and last-update timestamps.	No direct entity relationship.
Properties	IsDeleted	bool	Soft-delete flag in current baseline; confirm.	Retains product references/history for reports and inquiries.
Behaviour	UpdateDetails(...)	Method	Updates editable listing details.	Seller authorization is checked by Application use case.
Behaviour	ChangeStatus(...)	Method	Changes listing state through allowed transitions.	Uses ProductStatus enum.

Category

Responsibility: Groups products for browsing and filtering.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	Id	Guid	Unique category identifier.	Referenced by Product.CategoryId.
Properties	Name	string	Required category name.	One Category has zero or many Products: Category 1 : 0..* Product.
Properties	Description	string?	Optional category description.	No direct entity relationship.
Properties	Products	Collection<Product>	Products in the category, if navigation property is used.	One-to-many with Product; can be represented by FK without navigation collection.
Behaviour	Rename(...)	Method	Changes category name after validation.	Category management use case only if included in final requirements.

Report

Responsibility: A report submitted about a product or optionally a user.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	Id	Guid	Unique report identifier.	Referenced by admin review workflows.
Properties	ReporterId	Guid	User who submitted the report.	Many Reports belong to one ApplicationUser: ApplicationUser 1 : 0..* Report.
Properties	TargetProductId	Guid?	Product being reported; required by baseline.	Many Reports may target one Product: Product 1 : 0..* Report.
Properties	TargetUserId	Guid?	Optional user target; confirm requirement.	If enabled, many Reports may target one ApplicationUser: ApplicationUser 1 : 0..* Report.
Properties	Reason	string	Required reason for report.	No direct entity relationship.
Properties	Status	ReportStatus	Pending, Reviewed, or other agreed values.	Uses ReportStatus enum.
Properties	CreatedAt	DateTime	UTC submission timestamp.	No direct entity relationship.
Behaviour	Review(...)	Method	Records moderation decision/status.	Admin workflow; exactly one target should be set if user reporting is supported.

ContactInquiry

Responsibility: A buyer-to-seller inquiry about a product; not a multi-message chat.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	Id	Guid	Unique inquiry identifier.	Referenced by inquiry service and response DTOs.
Properties	ProductId	Guid	Product being queried.	Many inquiries may reference one Product: Product 1 : 0..* ContactInquiry.
Properties	BuyerId	Guid	Student who sends inquiry.	Many inquiries may belong to one buyer: ApplicationUser 1 : 0..* ContactInquiry.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	SellerId	Guid	Product owner / recipient.	Many inquiries may belong to one seller: ApplicationUser 1 : 0..* ContactInquiry.
Properties	Message	string	Initial inquiry message.	No direct entity relationship.
Properties	CreatedAt	DateTime	UTC creation timestamp.	No direct entity relationship.
Behaviour	ValidateParticipants()	Method	Ensures buyer/seller are valid and seller owns the product.	Coordinates Product and ApplicationUser references.

Domain Layer — Enums

UserRole

Responsibility: Supported user roles.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	Student	enum value	Standard marketplace user.	Used by ApplicationUser.Role and authorization.
Properties	Admin	enum value	Administrator role.	Used by ApplicationUser.Role and admin authorization.

ProductStatus

Responsibility: Lifecycle state for product listings.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	Available	enum value	Listing is available.	Used by Product.Status and search filters.
Properties	Sold	enum value	Listing has been sold.	Used by Product.Status.
Properties	Removed	enum value	Listing is removed from public view.	Used by Product.Status.
Properties	Archived	enum value	Optional archived state; confirm.	Used by Product.Status if approved.

ReportStatus

Responsibility: Lifecycle state for reports.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	Pending	enum value	Awaiting moderation.	Used by Report.Status.
Properties	Reviewed	enum value	Reviewed by admin.	Used by Report.Status; additional outcomes need agreement.

Application Layer — Service Contracts

IAuthService

Responsibility: Authentication use cases.

Section	Property / operation	Type	Details / responsibility	Relation
Behaviour	RegisterAsync(RegisterStudentDto)	Operation	Registers a student and returns a safe result.	Uses RegisterStudentDto and AuthResponseDto; delegates identity operations to IIdentityService.
Behaviour	LoginAsync(LoginDto)	Operation	Authenticates credentials without exposing secrets.	Uses LoginDto and AuthResponseDto; depends on IIdentityService.
Behaviour	LogoutAsync()	Operation	Ends the current authentication session.	Implemented through identity/authentication adapter.

IProfileService

Responsibility: Profile read and update use cases.

Section	Property / operation	Type	Details / responsibility	Relation
Behaviour	GetProfileAsync(userId)	Operation	Returns the current user's profile.	Reads ApplicationUser via IUserRepository and maps to UserProfileDto.
Behaviour	UpdateProfileAsync(userId, dto)	Operation	Validates and updates editable profile fields.	Uses UpdateProfileDto, ApplicationUser, IUserRepository, and UserProfileDto.

IProductService

Responsibility: Product listing and search use cases.

Section	Property / operation	Type	Details / responsibility	Relation
Behaviour	CreateAsync(sellerId, dto)	Operation	Validates and creates a listing.	Uses CreateProductDto, Product, IProductRepository, ICategoryRepository.
Behaviour	UpdateAsync(productId, actorId, dto)	Operation	Updates listing only for an authorized seller/admin.	Uses UpdateProductDto, Product, IProductRepository.
Behaviour	GetDetailsAsync(productId)	Operation	Returns product details.	Maps Product and related seller/category data to ProductDetailsDto.
Behaviour	SearchAsync(filter)	Operation	Searches and pages listings.	Uses ProductSearchFilterDto, IProductRepository, PagedResult<ProductSummaryDto>.

IInquiryService

Responsibility: Buyer-to-seller inquiry use cases.

Section	Property / operation	Type	Details / responsibility	Relation
Behaviour	CreateAsync(buyerId, dto)	Operation	Validates participants and product ownership.	Uses CreateInquiryDto, Product, ApplicationUser, IContactInquiryRepository.
Behaviour	GetForUserAsync(userId)	Operation	Returns inquiries visible to a participant.	Maps ContactInquiry to InquiryResponseDto; restricts to buyer/seller.

IAdminService

Responsibility: Administrative moderation and management use cases.

Section	Property / operation	Type	Details / responsibility	Relation
Behaviour	SearchUsersAsync(filter)	Operation	Searches users for admin review.	Uses UserSearchFilterDto, IUserRepository, AdminUserDto.
Behaviour	ManageProductAsync(...)	Operation	Performs approved admin product actions.	Uses ProductAdminFilterDto/AdminProductDto and IProductRepository.
Behaviour	ReviewReportAsync(reportId, decision)	Operation	Reviews a report and records outcome.	Uses Report, IReportRepository, ReportStatus.

Application Layer — DTOs and Results

RegisterStudentDto

Responsibility: Registration input.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	FullName / Email / CollegeName	string	Required registration fields.	Consumed by IAuthService; maps into ApplicationUser and identity record.
Properties	Password	string	Input credential; never returned in responses.	Passed only to IIdentityService; never persisted in Domain.

LoginDto

Responsibility: Login input.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	Email	string	User's login identifier.	Consumed by IAuthService / IIdentityService.
Properties	Password	string	Input credential.	Never logged or returned.

AuthResponseDto

Responsibility: Safe authentication response.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	UserId / FullName / Role	values	Non-secret identity information.	Returned by IAuthService; Role corresponds to UserRole.
Properties	Success / Message	values	Operation result and user-safe message.	Returned to AuthController; no password or hash.

UserProfileDto

Responsibility: Read-only profile response.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	Id / FullName / Email / CollegeName	values	Profile display data.	Mapped from ApplicationUser by IProfileService.
Properties	PhoneNumber / ProfileImageUrl	optional values	Optional profile display data.	Mapped from ApplicationUser.

UpdateProfileDto

Responsibility: Profile update input.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	FullName / PhoneNumber / CollegeName / ProfileImageUrl	values	Fields allowed to be edited.	Consumed by IProfileService; must not accept Role, Id, or CreatedAt.

CreateProductDto

Responsibility: Product creation input.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	Title / Description / Price / ImageUrl / CategoryId	values	Listing input; validate required fields and price > 0.	Consumed by IProductService; maps to Product; seller ID comes from authenticated actor.

UpdateProductDto

Responsibility: Product update input.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	Title / Description / Price / ImageUrl / CategoryId	values	Editable listing fields.	Consumed by IProductService; seller ownership must be checked.

ProductSearchFilterDto

Responsibility: Product search criteria.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	SearchTerm / CategoryId / MinPrice / MaxPrice / Status / Page	values	Optional filters and paging values.	Consumed by IProductService; CategoryId relates to Category and Status to ProductStatus.

ProductSummaryDto

Responsibility: Compact listing result.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	Id / Title / Price / ImageUrl / Status	values	Fields for product cards.	Mapped from Product; returned in search/listing results.

ProductDetailsDto

Responsibility: Full listing response.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	Product fields + Seller display data + CategoryName	values	Details view data without persistence types.	Mapped from Product, ApplicationUser, and Category by IProductService.

CreateReportDto

Responsibility: Report submission input.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	TargetProductId / TargetUserId / Reason	values	Target fields and reason; exactly one target if optional user reports are supported.	Consumed by IAdminService/report use case; maps to Report and target entities.

ReportResponseDto

Responsibility: Safe report response.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	Id / Target / Reason / Status / CreatedAt	values	Report information shown to user/admin.	Mapped from Report; Status uses ReportStatus.

CreateInquiryDto

Responsibility: Inquiry creation input.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	ProductId / Message	values	Product reference and initial message.	Consumed by IInquiryService; buyer ID comes from authenticated actor.

InquiryResponseDto

Responsibility: Inquiry response.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	Id / ProductId / BuyerId / SellerId / Message / CreatedAt	values	Inquiry data visible to participants.	Mapped from ContactInquiry and related product/user display data.

AdminUserDto

Responsibility: Admin user summary.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	Id / FullName / Email / Role / IsBlocked	values	Non-secret fields for user administration.	Mapped from ApplicationUser; Role uses UserRole.

AdminProductDto

Responsibility: Admin listing summary.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	Id / Title / SellerId / CategoryId / Price / Status	values	Fields for product moderation.	Mapped from Product and optional seller/category display data.

UserSearchFilterDto

Responsibility: Admin user search criteria.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	SearchTerm / Role / IsBlocked / Page	values	Optional filters and paging.	Consumed by IAdminService; Role uses UserRole.

ProductAdminFilterDto

Responsibility: Admin product search criteria.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	SearchTerm / CategoryId / Status / Page	values	Optional admin filters.	Consumed by IAdminService; CategoryId relates to Category and Status to ProductStatus.

PagedResult

Responsibility: Generic paged result wrapper.

Section	Property / operation	Type	Details / responsibility	Relation
Properties	Items / TotalCount / Page / PageSize	values	Result records and paging metadata.	Wraps DTO collections such as ProductSummaryDto or admin DTOs; generic T is not a domain relationship.

Application Layer — Repository / Identity Interfaces

IProductRepository

Responsibility: Product persistence abstraction.

Section	Property / operation	Type	Details / responsibility	Relation
Behaviour	GetByIdAsync / AddAsync / UpdateAsync	Operations	Retrieve and persist Product entities.	Implemented by Infrastructure ProductRepository; used by IProductService and admin use cases.
Behaviour	SearchAsync(filter)	Operation	Returns products matching filters.	Uses ProductSearchFilterDto or repository-specific criteria; returns Product entities.

ICategoryRepository

Responsibility: Category persistence abstraction.

Section	Property / operation	Type	Details / responsibility	Relation
Behaviour	GetByIdAsync / GetAllAsync	Operations	Retrieve categories.	Implemented by CategoryRepository; used by IProductService and Product.CategoryId validation.

IUserRepository

Responsibility: User profile persistence abstraction.

Section	Property / operation	Type	Details / responsibility	Relation
Behaviour	GetByIdAsync / SearchAsync / UpdateAsync	Operations	Retrieve/update ApplicationUser profiles.	Implemented by UserRepository; used by IProfileService and IAdminService.

IReportRepository

Responsibility: Report persistence abstraction.

Section	Property / operation	Type	Details / responsibility	Relation
Behaviour	AddAsync / GetByIdAsync / UpdateAsync	Operations	Create and review reports.	Implemented by ReportRepository; used by report/admin use cases.

IContactInquiryRepository

Responsibility: Inquiry persistence abstraction.

Section	Property / operation	Type	Details / responsibility	Relation
Behaviour	AddAsync / GetForUserAsync	Operations	Create and retrieve participant inquiries.	Implemented by ContactInquiryRepository; used by IInquiryService.

IIdentityService

Responsibility: Authentication provider boundary.

Section	Property / operation	Type	Details / responsibility	Relation
Behaviour	RegisterAsync / ValidateCredentialsAsync / SignOutAsync	Operations	Wraps ASP.NET Core Identity or chosen authentication mechanism.	Implemented in Infrastructure; used by IAuthService; maps identity IDs to ApplicationUser.Id.

Infrastructure and WebUI

ProductRepository / CategoryRepository / UserRepository / ReportRepository / ContactInquiryRepository

Responsibility: EF Core repository implementations.

Section	Property / operation	Type	Details / responsibility	Relation
Behaviour	Repository operations	Methods	Implement Application repository interfaces using persistence.	Each implementation maps one repository contract to CampusXDbContext and corresponding Domain entity; Infrastructure depends inward on Application and Domain.

IdentityService

Responsibility: Identity adapter implementation.

Section	Property / operation	Type	Details / responsibility	Relation
Behaviour	Register / Login / Logout	Methods	Uses ASP.NET Core Identity or selected provider.	Implements IIdentityService; maps identity user ID to ApplicationUser.Id; credentials remain outside Domain.

AuthController

Responsibility: MVC endpoints for authentication.

Section	Property / operation	Type	Details / responsibility	Relation
Behaviour	Register / Login / Logout actions	Actions	Accept ViewModels, invoke IAuthService, return views/redirects.	Depends on IAuthService; no direct database access.

ProfileController

Responsibility: MVC endpoints for profile management.

Section	Property / operation	Type	Details / responsibility	Relation
Behaviour	Details / Edit actions	Actions	Display and update current user's profile.	Depends on IProfileService; actor ID comes from authenticated principal.

ProductsController

Responsibility: MVC endpoints for product listings.

Section	Property / operation	Type	Details / responsibility	Relation
Behaviour	Index / Details / Create / Edit actions	Actions	Browse and manage product listings.	Depends on IProductService; service uses repositories and Domain entities.

InquiriesController

Responsibility: MVC endpoints for inquiries.

Section	Property / operation	Type	Details / responsibility	Relation
Behaviour	Create / MyInquiries actions	Actions	Create inquiries and display participant-specific records.	Depends on IInquiryService; access scoped to buyer/seller.

ReportsController

Responsibility: MVC endpoints for report submission.

Section	Property / operation	Type	Details / responsibility	Relation
Behaviour	Create action	Action	Accepts report submission and returns safe result.	Depends on report use case/service; maps to Report.

AdminUsersController / AdminProductsController / AdminReportsController

Responsibility: Admin-only MVC endpoints.

Section	Property / operation	Type	Details / responsibility	Relation
Behaviour	Search / Manage / Review actions	Actions	Perform authorized administrative operations.	Depend on IAdminService; admin policy required.

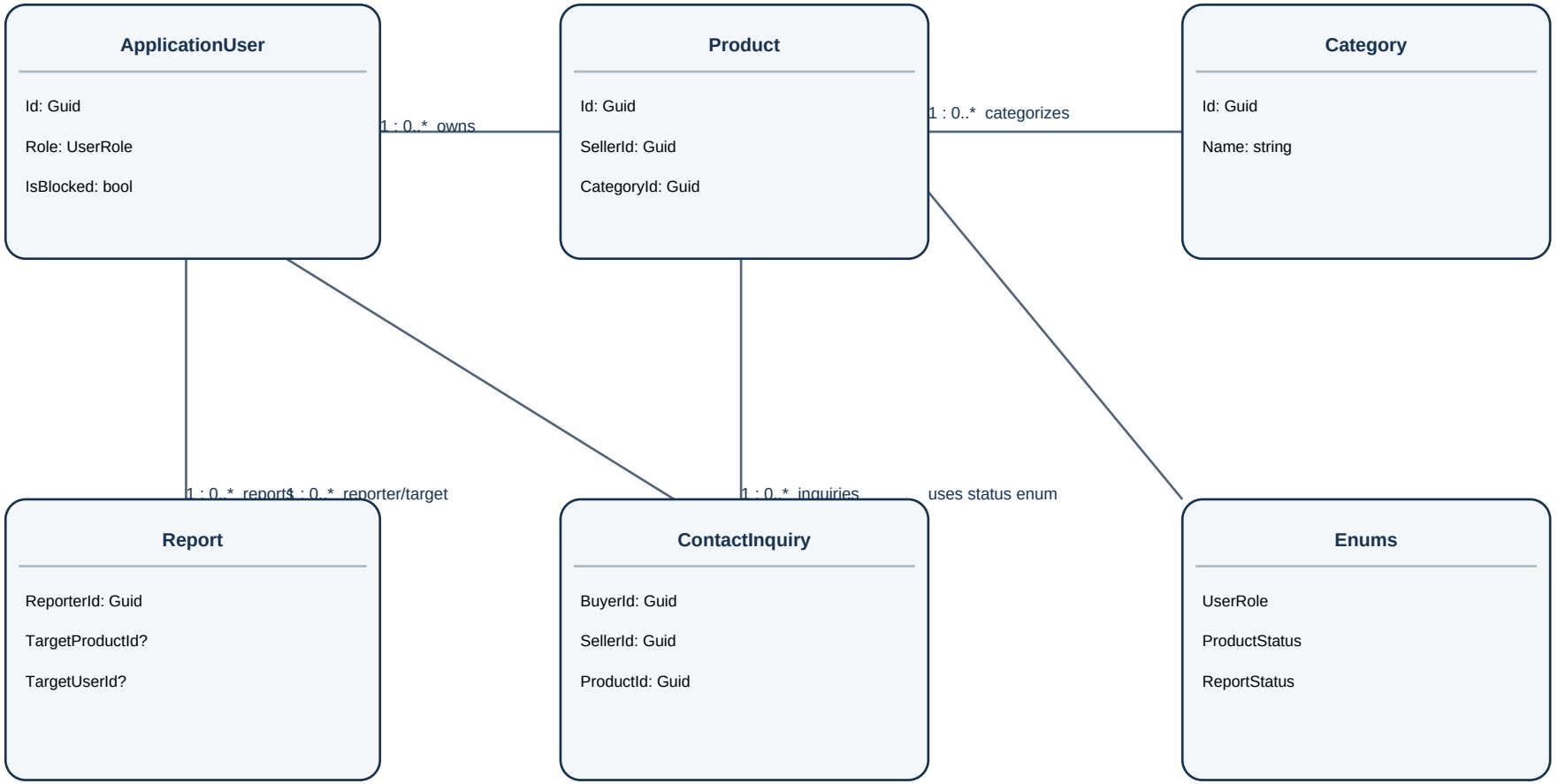
Dependency Injection Configuration

Responsibility: WebUI composition root.

Section	Property / operation	Type	Details / responsibility	Relation
Behaviour	Service registrations	Configuration	Maps Application interfaces to Infrastructure implementations.	Connects WebUI to Application contracts and Infrastructure implementations; composition root is the only inward-to-outward wiring point.

UML Class Diagram — Domain Relationships

The diagram focuses on the core Domain entities and enum dependencies. Multiplicities reflect the current specification baseline; optional user reporting must be confirmed.



Clean Architecture dependency direction: WebUI → Application → Domain. Infrastructure implements Application abstractions and depends inward on Application/Domain. Domain has no dependency on outer layers.

Implementation Decisions to Confirm

Decision	Current baseline	Confirm before implementation
Identity	Domain ApplicationUser is separate from ASP.NET Core Identity user.	Confirm Identity setup and ID mapping.
Product deletion	IsDeleted soft deletion is assumed.	Confirm soft delete vs hard delete and history retention.
Product image	ImageUrl is treated as required based on the current specification baseline.	Confirm upload/storage and validation requirements.
Report target	Product reporting is required; user reporting is optional.	Confirm whether students can report users; exactly one target should be set.
Statuses	Available/Sold/Removed/Archived and report statuses are listed as baseline values.	Confirm allowed values and transitions.
Persistence	Repository commit or Unit of Work strategy must be consistent.	Choose one transaction strategy.

CampusX — Relationship Multiplicity Reference

Supplement to the Classes Specifications Document • Peer-to-Peer College Student Marketplace

How to read the notation

1:M means one-to-many. **1:0..*** means one-to-zero-or-many. **M:1** is the reverse view of one-to-many. These are conceptual relationships in the current specification baseline; confirm optional reporting and final persistence choices with the team.

Domain entity relationships

Class	Relation	Multiplicity / meaning
ApplicationUser	Product	1 : 0..* — one user may list zero or many products; each product has one seller.
Category	Product	1 : 0..* — one category may contain zero or many products; each product references one category.
ApplicationUser	Report (Reporter)	1 : 0..* — one user may submit zero or many reports; each report has one reporter.
Product	Report (TargetProduct)	1 : 0..* — one product may be targeted by zero or many reports; each product report references one product.
ApplicationUser	Report (TargetUser)	1 : 0..* — optional relationship, only if reporting a user is supported. TargetUserId is nullable.
Product	ContactInquiry	1 : 0..* — one product may receive zero or many inquiries; each inquiry references one product.
ApplicationUser	ContactInquiry (Buyer)	1 : 0..* — one user may send zero or many inquiries; each inquiry has one buyer.
ApplicationUser	ContactInquiry (Seller)	1 : 0..* — one user may receive zero or many inquiries as seller; each inquiry has one seller.

Important: The multiplicity is about the conceptual model, not a requirement to add navigation properties in EF Core. Foreign keys can represent these relationships without collection/navigation properties. Do not add a many-to-many relationship unless the SRS and implementation require it.

How the Relation column should be written

Use short, explicit multiplicity notation for entity-to-entity relationships. Keep implementation notes and assumptions in Details / responsibility.

Examples for the class tables

Property / operation	Relation column
ApplicationUser.Id	Product.SellerId — 1:M from ApplicationUser to Product; M:1 from Product to ApplicationUser.
Product.CategoryId	Category 1:M Product; Product M:1 Category.
Report.ReporterId	ApplicationUser 1:M Report; each Report has one Reporter.
Report.TargetProductId	Product 1:M Report; each product-targeted Report has one Product.
Report.TargetUserId	ApplicationUser 1:M Report (optional; only if user reporting is enabled).
ContactInquiry.ProductId	Product 1:M ContactInquiry; each ContactInquiry has one Product.
ContactInquiry.BuyerId	ApplicationUser 1:M ContactInquiry (Buyer role).
ContactInquiry.SellerId	ApplicationUser 1:M ContactInquiry (Seller role).

For DTOs, service contracts, repositories, controllers, and enum types, the Relation column should describe dependency/usage rather than database cardinality. For example: “IProductRepository depends on Product entity” or “Product.Status uses ProductStatus enum.” Do not assign 1:M to a service or DTO merely because it references an entity.

Clean Architecture reminder

Domain entities and enums belong to Domain. Application services, use cases, DTOs, and repository abstractions depend inward on Domain. Infrastructure implements repository abstractions. WebUI depends on Application for use cases and contracts. These layer dependencies are not entity multiplicities and should be documented separately.